



Science

Stage 9

Paper 2

2022

Cambridge Lower Secondary Progression Test

Name

Class

Date

45 minutes

No additional materials are needed.

INSTRUCTIONS

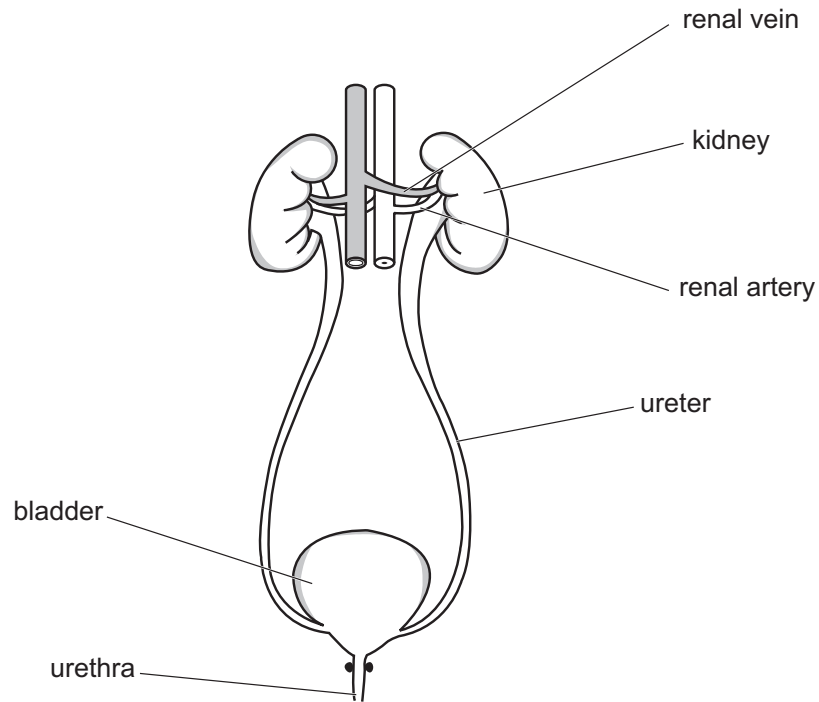
- Answer **all** questions.
- Write your answer to each question in the space provided.
- You should show all your working on the question paper.

INFORMATION

- The total mark for this paper is 50.
- The number of marks for each question or part question is shown in brackets [].

1 Look at the diagram.

It shows the human excretory (renal) system.



(a) Look at the table about what happens in different parts of the excretory (renal) system.

Complete the table.

The first row has been done for you.

part	what happens
renal artery	blood enters excretory (renal) system
kidney	
.....	urine travels through this tube to be stored
.....	urine is stored here before excretion from body

[3]

- (b) Kidney failure causes a build-up of a toxic substance in the blood.

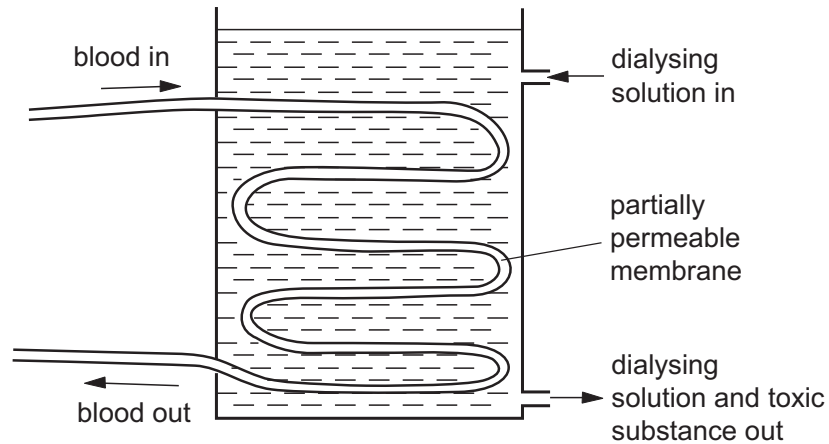
Write down the name of this toxic substance.

..... [1]

- (c) Artificial kidneys prevent the build-up of toxic substances in the blood.

Artificial kidneys use a process called dialysis.

Look at the diagram of an artificial kidney.



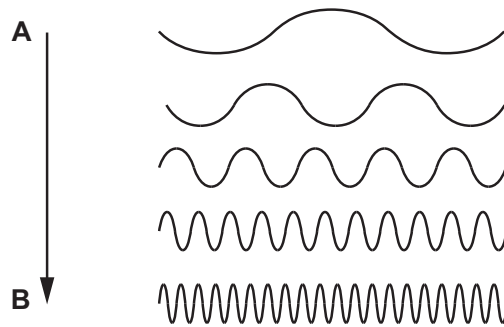
Use the model to describe how dialysis works.

.....

 [2]

2 Look at the diagram.

It shows different sound traces on an oscilloscope.



(a) Complete the sentences to describe what is happening to the sound from **A** to **B**.

Choose from the list.

decreases

increases

stays the same

The loudness of the sound

The frequency of the sound

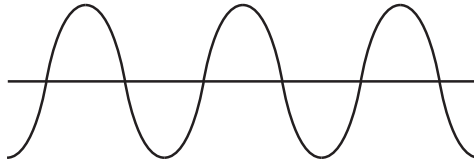
[2]

(b) This question is about combining sound waveforms.

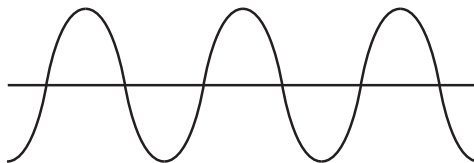
The diagram shows two waveforms, wave **1** and wave **2**, that combine together.

It also shows the new waveform made.

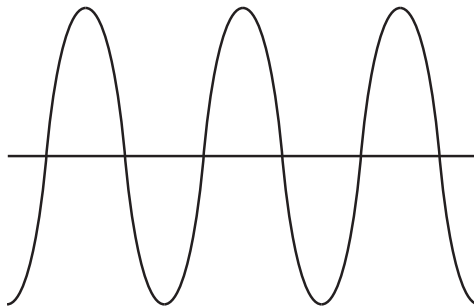
wave **1**



wave **2**



wave **1** + wave **2**

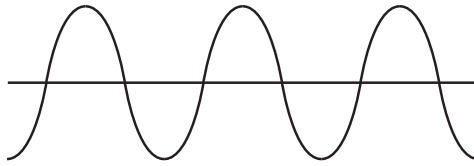


(i) Write down the word that describes how wave **1** and wave **2** combine together.

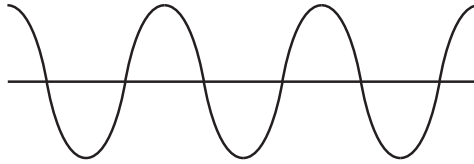
..... [1]

- (ii) Two more waveforms, wave **3** and wave **4**, combine together.

wave **3**



wave **4**



wave **3** + wave **4**

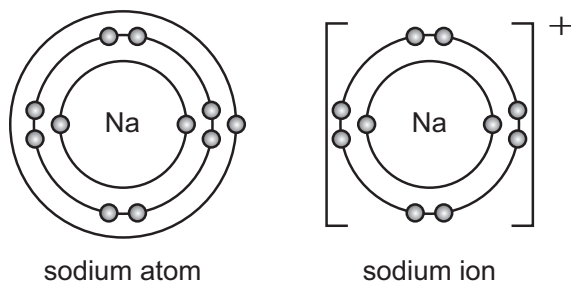


Write down the word that describes how wave **3** and wave **4** combine together.

..... [1]

3 Look at the diagrams.

They show the electronic structures of a sodium atom and of a sodium ion.

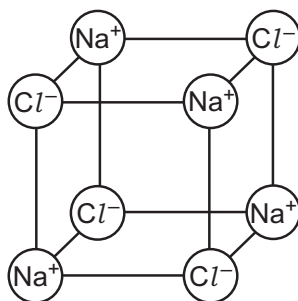


(a) Describe how a sodium ion is made from a sodium atom.

..... [1]

(b) Look at the diagram.

It shows the structure of sodium chloride.



(i) Write down the name of the **type** of bonding in sodium chloride.

..... [1]

(ii) The bonding between sodium ions and chloride ions is strong.

Explain why.

..... [1]

(iii) Sodium chloride has a giant structure.

Circle the melting point of sodium chloride.

–50 °C

0 °C

52 °C

801 °C

[1]

4 This question is about tectonic plates.

(a) Tectonic plates float on molten magma in the mantle.

(i) Write down the name of the layer of the Earth made of tectonic plates.

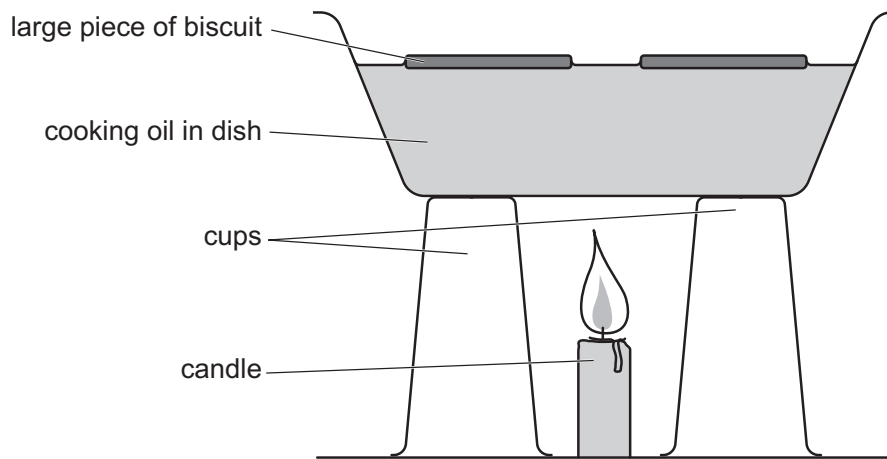
..... [1]

(ii) Explain why tectonic plates float on the mantle.

..... [1]

(b) Yuri models what happens at the boundary between two tectonic plates.

Look at the diagram of Yuri's model.



Each large piece of biscuit models a tectonic plate.

The large pieces of biscuit float on the cooking oil.

The cooking oil models the magma in the mantle.

(i) Which part of the Earth does the lighted candle in the model represent?

..... [1]

(ii) Predict what will happen to the pieces of biscuit.

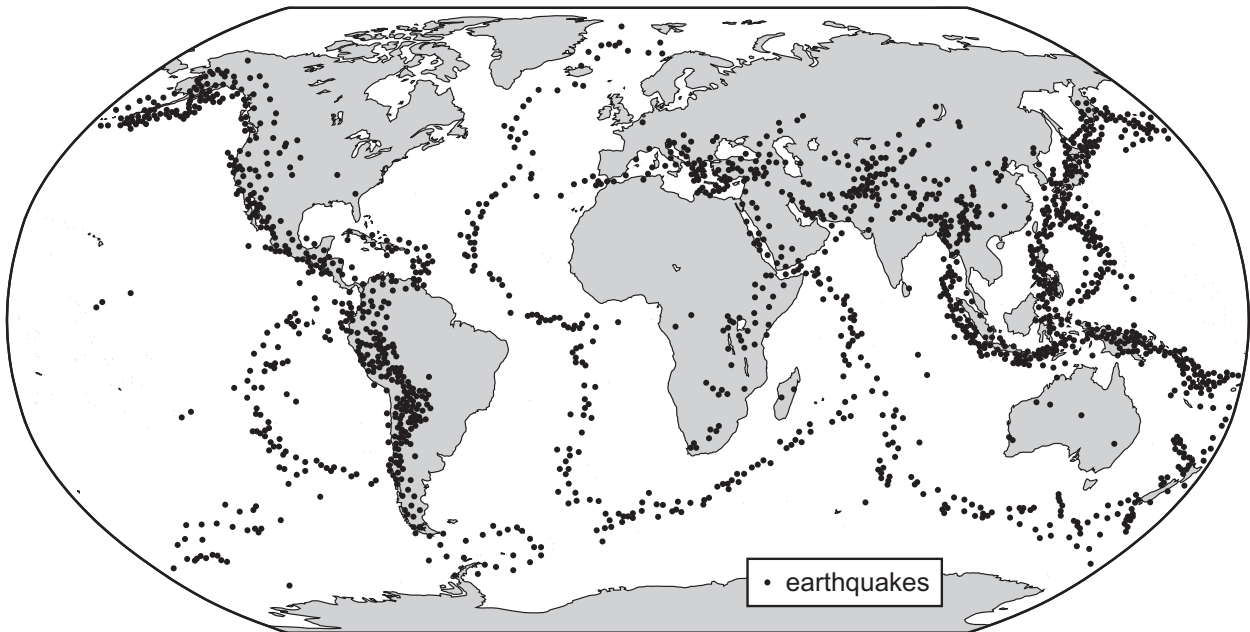
Explain your answer.

prediction

explanation

..... [2]

(iii) The map shows the positions of earthquakes on the Earth's surface.



Explain how the positions of earthquakes are used as evidence for tectonic plates.

.....

.....

..... [2]

(iv) Write down **two other** features that are used as evidence for tectonic plates.

1

2 [2]

(c) There have been five times in the Earth's history when most of the organisms living became extinct. This is called a mass extinction.

Scientists think that enormous volcanic eruptions caused these mass extinctions.

Explain why.

.....

..... [1]

- 5 Priya finds some information about the effect of smoking on fetal development.

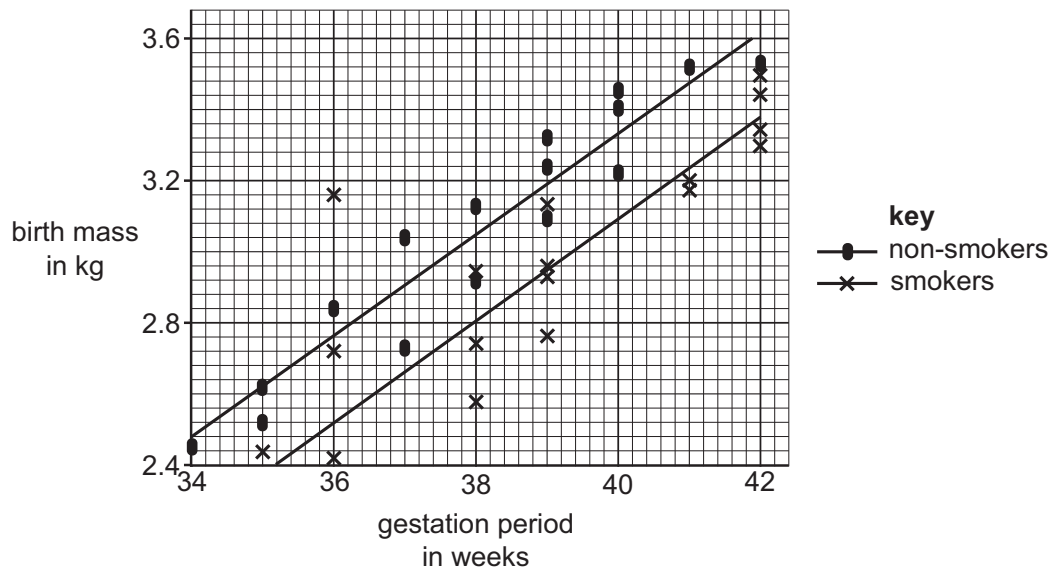
Look at the graph.

It shows the birth mass of:

- 16 babies born to non-smokers
- 16 babies born to smokers.

The graph also shows the gestation period.

This is the time period between fertilisation of an egg and the birth of a baby.



- (a) (i) Look at the birth masses for a gestation period of 36 weeks.

One of the birth masses is anomalous.

Circle the anomalous birth mass on the graph.

[1]

- (ii) Suggest **one** possible conclusion from the data.

..... [1]

- (b) Describe **one** way the evidence collected could be made more reliable.

.....
 [1]

- 6 Look at the information about Group 1 elements.

element	melting point in °C	boiling point in °C
sodium	98	883
potassium	64	759
rubidium	39	688

- (a) Lithium is **above** sodium in the Periodic Table.

Predict the melting point of lithium.

..... °C [1]

- (b) Caesium is **below** rubidium in the Periodic Table.

Predict the boiling point of caesium.

..... °C [1]

- (c) Which of the **three** elements in the table is the **most** reactive?

..... [1]

7 Aiko is in a hot room.

She starts to produce perspiration on her skin.

Perspiration is a watery liquid.

The perspiration helps Aiko to control her body temperature.

Complete the sentences to explain how.

Choose from the list.

boil	chemical	condense	cool
decreases	electrical	evaporate	increases
melts	potential	stays the same	thermal

The water in the perspiration begins to

During this process the water absorbs energy from the skin.

The temperature of the skin

[3]

- 8 This question is about inheritance in humans.

Reproduction is important for the survival of the species.

- (a) A fertilised human egg cell contains 46 chromosomes.

How many chromosomes are there in a sperm cell?

.....

Explain your answer.

.....

.....

[2]

- (b) Human sperm production is damaged by temperatures 2 °C above normal body temperature.

This could affect the survival of the human species.

Suggest how an increase in the temperature of the environment affects the ability of humans to reproduce.

.....

.....

.....

.....

[2]

- (c) Write down the word that describes the total number of individuals of the same species.

.....

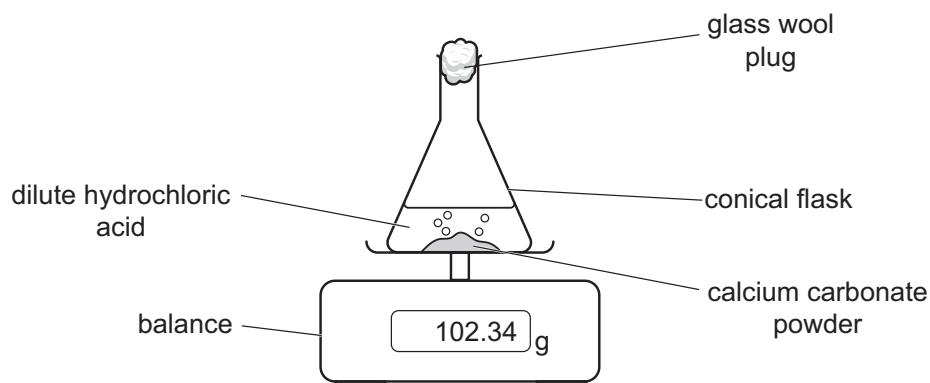
[1]

- 9 Chen investigates the rate of reaction between calcium carbonate and dilute hydrochloric acid.

Look at the word equation for the reaction.



Look at the equipment Chen uses.



- (a) Predict what happens to the mass reading on the balance during the reaction.

.....

Explain your answer.

.....

.....

[2]

- (b) Chen finds that the rate of the reaction is greater when he uses powdered calcium carbonate rather than lumps of calcium carbonate.

Explain why using the particle model.

.....

.....

.....

[2]

- (c) Describe **one** safety precaution Chen must take during his investigation.

.....

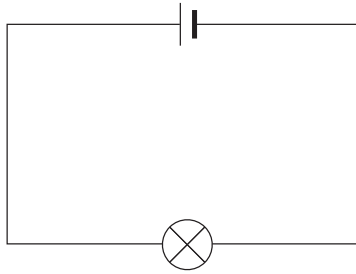
Explain why he takes this safety precaution.

.....

[2]

10 Angelique wants to measure the current and voltage in a circuit.

(a) Look at the circuit diagram.



(i) Angelique measures the current using an ammeter.

Draw the position of the ammeter on the circuit diagram.

Use the correct symbol for the ammeter.

[1]

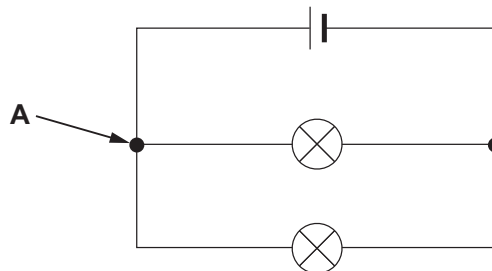
(ii) Angelique measures the voltage across the lamp using a voltmeter.

Draw the position of the voltmeter on the circuit diagram.

Use the correct symbol for the voltmeter.

[1]

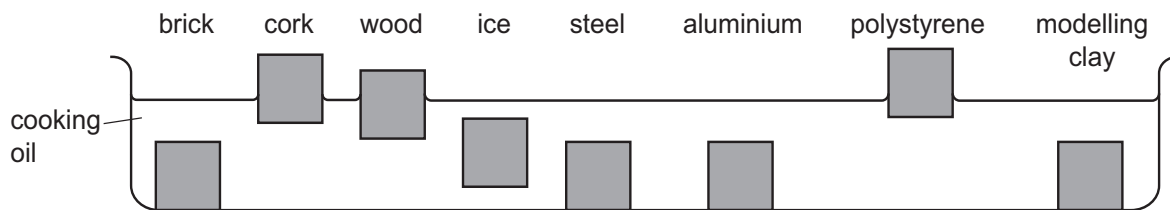
(b) Angelique builds another circuit.



Describe what happens to the current in the circuit at point **A**.

..... [1]

11 Hassan investigates if objects float or sink when placed in cooking oil.



Look at the table of densities.

material	density in g/cm^3
brick	2.1
cork	0.2
wood	0.7
ice	0.9
steel	8.1
aluminium	2.7
polystyrene	0.3
modelling clay	1.7
cooking oil	0.9

(a) Hassan uses the diagram and the data in the table to make a hypothesis about which objects float and which objects sink in cooking oil.

He wants to link the density of an object to if it floats or sinks in cooking oil.

Suggest the hypothesis Hassan makes.

.....

.....

..... [1]

(b) Hassan investigates if objects made from plastics of different densities will float in **water**.

Describe his investigation.

.....

.....

.....

..... [2]

The Periodic Table of Elements

Group																	
1	2	1 H hydrogen 1						7	8								
3 Li lithium 7	4 Be beryllium 9	Key atomic number atomic symbol name relative atomic mass						9 F fluorine 19	10 Ne neon 20								
11 Na sodium 23	12 Mg magnesium 24							16 S sulfur 32	18 Ar argon 40								
19 K potassium 39	20 Ca calcium 40	21 Sc scandium 45	22 Ti titanium 48	23 V vanadium 51	24 Cr chromium 52	25 Mn manganese 55	26 Fe iron 56	27 Co cobalt 59	28 Ni nickel 58	29 Cu copper 64	30 Zn zinc 65	31 Ga gallium 70	32 Ge germanium 73	33 As arsenic 75	34 Se selenium 79	35 Br bromine 80	36 Kr krypton 84
37 Rb rubidium 85	38 Sr strontium 88	39 Y yttrium 89	40 Zr zirconium 91	41 Nb niobium 93	42 Mo molybdenum 96	43 Tc technetium —	44 Ru ruthenium 101	45 Rh rhodium 103	46 Pd palladium 106	47 Ag silver 108	48 Cd cadmium 112	49 In indium 115	50 Sn tin 119	51 Sb antimony 122	52 Te tellurium 128	53 I iodine 127	54 Xe xenon 131
55 Cs caesium 133	56 Ba barium 137	57–71 lanthanoids	72 Hf hafnium 178	73 Ta tantalum 181	74 W tungsten 184	75 Re rhenium 186	76 Os osmium 190	77 Ir iridium 192	78 Pt platinum 195	79 Au gold 197	80 Hg mercury 201	81 Tl thallium 204	82 Pb lead 207	83 Bi bismuth 209	84 Po polonium —	85 At astatine —	86 Rn radon —
87 Fr francium —	88 Ra radium —	89–103 actinoids	104 Rf rutherfordium —	105 Db dubnium —	106 Sg seaborgium —	107 Bh bohrium —	108 Hs hassium —	109 Mt meitnerium —	110 Ds darmstadtium —	111 Rg roentgenium —	112 Cn copernicium —	113 Nh nihonium —	114 Fl flerovium —	115 Mc moscovium —	116 Lv livermorium —	117 Ts tennessine —	118 Og oganesson —
lanthanoids																	
57 La lanthanum 139	58 Ce cerium 140	59 Pr praseodymium 141	60 Nd neodymium 144	61 Pm promethium —	62 Sm samarium 150	63 Eu europium 152	64 Gd gadolinium 157	65 Tb terbium 159	66 Dy dysprosium 163	67 Ho holmium 165	68 Er erbium 167	69 Tm thulium 169	70 Yb ytterbium 173	71 Lu lutetium 175			
89 Ac actinium	90 Th thorium 232	91 Pa protactinium 231	92 U uranium 238	93 Np neptunium —	94 Pu plutonium —	95 Am americium —	96 Cm curium —	97 Bk berkelium —	98 Cf californium —	99 Es einsteinium —	100 Fm fermium —	101 Md mendelevium —	102 No nobelium —	103 Lr lawrencium —			
actinoids																	

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